CS216-BLOCKCHAIN TRANSACTION ASSIGNMENT

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A.Objective of the Assignment:

The objective of this assignment is to understand the process of creating and validating Bitcoin transactions using Legacy (P2PKH) and SegWit (P2SH-P2WPKH) address formats. We interact with bitcoind using Python, create raw transactions, decode scripts, and analyze how Bitcoin scripts validate transactions.

B.TOOLS AND ENVIRONMENT SETUP

B.1.TOOLS USED

The following tools were used to perform the Bitcoin transaction experiments:

Bitcoin Core (bitcoind)-

Bitcoin Core is the reference implementation of the Bitcoin protocol. It runs a full Bitcoin node and is responsible for maintaining the blockchain, validating transactions, and managing wallets. The bitcoind daemon runs in the background and allows programs to interact with the Bitcoin network.

Bitcoin CLI (bitcoin-cli)-

bitcoin-cli is a command-line interface used to interact with the Bitcoin daemon. It allows execution of various Remote Procedure Call (RPC) commands such as generating addresses, creating transactions, mining blocks, and querying blockchain data.

Python-

Python was used to write scripts that interact with the Bitcoin node using RPC calls. The assignment programs were implemented in Python to automate transaction creation, signing, and broadcasting.

python-bitcoinlib-

python-bitcoinlib is a Python library used to construct and manipulate Bitcoin transactions programmatically. It provides utilities for handling scripts, transactions, and Bitcoin data structures.

Bitcoin Script Debugger (btcdeb)-

btcdeb is a debugging tool used to analyze and step through Bitcoin Script execution. It allows verification of the correctness of the locking and unlocking scripts used in transactions.

These tools were required to create, analyze, and validate Bitcoin transactions in both legacy and SegWit formats.

B.2.ENVIRONMENT SETUP

The experiments were conducted using a local Bitcoin testing environment called Regtest mode.

Regtest mode is a private blockchain environment where blocks can be mined instantly and transactions can be tested without using real cryptocurrency. This provides a controlled environment for experimentation and debugging.

PART-1: LEGACY TRANSACTION

Adress Generation:

Three legacy addresses were genereated for the transaction chain and are shown in the screenshot below.

```
==================== Legacy Addresses ====================
{
  "A": "mh1dfDevNDv8d9oPTXu2xANJ9Kh5Ga1poq",
  "B": "n1c5pPaBdJ7Njou41bQzuLhLWtHDNvHYhDb",
  "C": "mrrSdGDLiem8dTv97LSnZpcf9vz65sofqq",
  "address_type": "legacy"
}
==========================================================
```

FUNDING ADDRESS A:

Address A was funded with an initial balance of 100.0 BTC using the sendtoaddress RPC command. This created the initial Unspent Transaction Output (UTXO) required to begin the transaction chain.

```
==================== Funding A ====================
{
  "funding_txid": "32bed9fb2eb78be012a79af3b8c85b240d21a3d0a921e9e5d96472249e921954",
  "funded_address": "mh1dfDevNDv8d9oPTXu2xANJ9Kh5Ga1poq",
  "fund_amount": 0.01,
  "confirmation_blocks_mined": 1,
  "mining_address": "bcrt1qkfjgd3w2h56l65wc7udypgauka5499kfa5q9uv"
}
=====================================================
```

TRANSACTION A->B:

Input: The UTXO from Address A.

Outputs: the BTC sent to Address B , with the remaining balance minus fees sent back to Address A as change.

TXID:
89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05


```
================= Legacy A -> B Artifact =================
{
  "from_address": "mh1dfDevNDvBd9oPTXu2xANL9Mh5Galpoq",
  "to_address": "n1c5pPaB6J7Njou4lbQzUhLWtHDNvHYhDb",
  "requested_amount": 0.004,
  "selected_utxo": {
    "txid": "32bed9fb2ab78be012a79af3b8c85b240bd21a3d0a921e9e5d964722490921954",
    "vout": 1,
    "address": "mh1dfDevNDvBd9oPTXu2xANL9Mh5Galpoq",
    "amount": 0.01,
    "confirmations": 1,
    "spendable": true,
    "solvable": true,
    "desc": "pkh([aa154e90/44h/1h/0h/0/1]03c423f233f957f9144dc4383f0924fce9c56e409dc02185b92abfdab2e15cfc7a)#x64ur44g",
  },
  "raw_tx_hex": "02000000015419929e247264d9e5e921a9d0a3210d245bc8b8f39aa712e08bb72efbd9be320100000000fdffffff01801a060000000000001976a914dc5ae01a049516a1c1c72d913d890243869a8c1f88ac00000000",
  "funded_tx": {
    "hex": "02000000015419929e247264d9e5e921a9d0a3210d245bc8b8f39aa712e08bb72efbd9be320100000000fdffffff02801a060000000000001976a914dc5ae01a049516a1c1c72d913d890243869a8c1f88acdf2609000000000001976a91490ff3a3ba1af01a8391dd509e211856388889 4a188ac00000000",
    "fee": 2.25e-06,
    "changepos": 1
  },
  "signed_tx_hex": "02000000015419929e247264d9e5e921a9d0a3210d245bc8b8f39aa712e08bb72efbd9be320100000006a47304402206c93c7c6a52d896664d7f21438729c34666886dddabffd74bab1ccc618e5cd9ec02205e93d0e814b067ce9f1d20a6ebe3378405e185f6bf58cb9517621add3a94073a012103c423f233f957f9144dc4383f0924fce9c56e409dc02185b92abfdab2e15cfc7afdffffff02801a060000000000001976a914dc5ae01a049516a1c1c72d913d890243869a8c1f88acdf2609000000000001976a91490ff3a3ba1af01a8391dd509e211856388889 4a188ac00000000",
  "txid": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
  "decoded_tx": {
    "txid": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
    "hash": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
    "version": 2,
    "size": 225,
    "vsize": 225,
    "weight": 900,
    "locktime": 0,
    "vin": [
      {
        "txid": "32bed9fb2ab78be012a79af3b8c85b240bd21a3d0a921e9e5d964722490921954",
        "vout": 1,
        "scriptSig": {
          "asm": "304402206c93c7c6a52d896664d7f21438729c34666886dddabffd74bab1ccc618e5cd9ec02205e93d0e814b067ce9f1d20a6ebe3378405e185f6bf58cb9517621add3a94073a[ALL] 03c423f233f957f9144dc4383f0924fce9c56e409dc02185b92abfdab2e15cfc7
a",
          "hex": "47304402206c93c7c6a52d896664d7f21438729c34666886dddabffd74bab1ccc618e5cd9ec02205e93d0e814b067ce9f1d20a6ebe3378405e185f6bf58cb9517621add3a94073a012103c423f233f957f9144dc4383f0924fce9c56e409dc02185b92abfdab2e15cfc7
a"
        },
        "sequence": 4294967293
      }
    ],
    "vout": [
      {
        "value": 0.004,
        "n": 0,
        "scriptPubKey": {
          "asm": "OP_DUP OP_HASH160 dc5ae01a049516a1c1c72d913d890243869a8c1f OP_EQUALVERIFY OP_CHECKSIG",
          "desc": "addr(n1c5pPaB6J7Njou4lbQzUhLWtHDNvHYhDb)#ynHs6r3u",
          "hex": "76a914dc5ae01a049516a1c1c72d913d890243869a8c1f88ac",
          "address": "n1c5pPaB6J7Njou4lbQzUhLWtHDNvHYhDb",
          "type": "pubkeyhash"
        }
      },
      {
        "value": 0.00599775,
        "n": 1,
        "scriptPubKey": {
          "asm": "OP_DUP OP_HASH160 90ff3a3ba1af01a8391dd509e211856388889 4a1 OP_EQUALVERIFY OP_CHECKSIG",
          "desc": "addr(mtjdMCARfMM2S1VDy2kCzaMtFGct9RUnY7)#m0lg6kx",
          "hex": "76a91490ff3a3ba1af01a8391dd509e211856388889 4a188ac",
          "address": "mtjdMCARfMM2S1VDy2kCzaMtFGct9RUnY7",
          "type": "pubkeyhash"
        }
      }
    ]
  },
  "tx_summary": {
    "txid": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
    "hash": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
    "version": 2,
    "size": 225,
    "vsize": 225,
    "weight": 900,
    "locktime": 0,
    "vin_count": 1,
    "vout_count": 2
  },
  "script_view": {
    "tx_summary": {
      "txid": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
      "hash": "89a53f5bb43846bf99f0869963c405751835a91734d064806471d800f378f05",
      "version": 2,
      "size": 225,
      "vsize": 225,
      "weight": 900,
      "locktime": 0,
      "vin_count": 1,
      "vout_count": 2
    },
    "inputs": [
      {
        "txid": "32bed9fb2ab78be012a79af3b8c85b240bd21a3d0a921e9e5d964722490921954",
        "vout": 1,
        "sequence": 4294967293,
        "scriptsig_asm": "304402206c93c7c6a52d896664d7f21438729c34666886dddabffd74bab1ccc618e5cd9ec02205e93d0e814b067ce9f1d20a6ebe3378405e185f6bf58cb9517621add3a94073a[ALL] 03c423f233f957f9144dc4383f0924fce9c56e409dc02185b92abfdab
2e15cfc7a",
        "scriptsig_hex": "47304402206c93c7c6a52d896664d7f21438729c34666886dddabffd74bab1ccc618e5cd9ec02205e93d0e814b067ce9f1d20a6ebe3378405e185f6bf58cb9517621add3a94073a012103c423f233f957f9144dc4383f0924fce9c56e409dc02185b92abfdab
2e15cfc7a",
        "txinwitness": []
      }
    ],
    "outputs": [
      {
        "value": 0.004,
        "n": 0,
        "scriptpubkey_type": "pubkeyhash",
        "scriptpubkey_asm": "OP_DUP OP_HASH160 dc5ae01a049516a1c1c72d913d890243869a8c1f OP_EQUALVERIFY OP_CHECKSIG",
        "scriptpubkey_hex": "76a914dc5ae01a049516a1c1c72d913d890243869a8c1f88ac",
        "address": "n1c5pPaB6J7Njou4lbQzUhLWtHDNvHYhDb",
        "addresses": []
      },
```


TRANSACTION B->C:

Input: To execute this transfer, the specific A -> B transaction was used as the input UTXO. This proves the continuous flow of funds.

Outputs: the BTC sent to Address C, with the remaining balance minus fees sent back to Address B as change.

txid:
ebe3be3abc08e8e153ad87d4a0e493e9e3e785bb29181703c3685147b2ec9a674

```
================= Legacy B -> C Artifact =================
{
  "from_address": "n1c5pPaBdJ7Njou41bQzuLhLWtHDNvHYhDb",
  "to_address": "mrrSdGDLiem8dTv97LSnZpcf9vz65sofqq",
  "requested_amount": 0.0025
```


```
================= Legacy B -> C Artifact =================
{
  "from_address": "n1c5pPaBdJ7Njou41bQzuLhLWtHDNvHYhDb",
  "to_address": "mrrSdGDLiem8dTv97LSnZpcf9vz65sofqq",
  "requested_amount": 0.0025,
  "selected_utxo": {
    "txid": "89a53f5b8b43846bf99f8869963c405751835a91734d064806471d800f378f05",
    "vout": 0,
    "address": "n1c5pPaBdJ7Njou41bQzuLhLWtHDNvHYhDb",
    "amount": 0.004,
    "confirmations": 1,
    "spendable": true,
    "solvable": true,
    "desc": "pkh([ea154e90/44h/1h/0h/0/4]027a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65)#rdjxunvv"
  },
  "raw_tx_hex": "020000000105bf370f801d4706480644d73915a835157403c9669089ff96b8443bb5b3fa589000000000dffffffff0190d003000000000001976a9147c58efc3e5a30cea3873e905e29ea4d9165d4d6d88ac00000000",
  "funded_tx": {
    "hex": "020000000105bf370f801d4706480644d73915a835157403c9669089ff96b8443bb5b3fa589000000000dffffffff0290d003000000000001976a9147c58efc3e5a30cea3873e905e29ea4d9165d4d6d88ac0f4902000000000001976a9148edb9697e77b2b7058d6b601c5d833e4011add fa88ac00000000",
    "fee": 2.25e-06,
    "changepos": 1
  },
  "signed_tx_hex": "020000000105bf370f801d4706480644d73915a835157403c9669089ff96b8443bb5b3fa589000000006a47304402200d3cdf2926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a230121027a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65ffffffff0290d003000000000001976a9147c58efc3e5a30cea3873e905e29ea4d9165d4d6d88ac0f4902000000000001976a9148edb9697e77b2b7058d6b601c5d833e4011addfa88ac00000000",
  "txid": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
  "decoded_tx": {
    "txid": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
    "hash": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
    "version": 2,
    "size": 225,
    "vsize": 225,
    "weight": 900,
    "locktime": 0,
    "vin": [
      {
        "txid": "89a53f5b8b43846bf99f8869963c405751835a91734d064806471d800f378f05",
        "vout": 0,
        "scriptSig": {
          "asm": "304402200d3cdf2926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a23[ALL] 027a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65",
          "hex": "47304402200d3cdf2926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a230121027a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65"
        },
        "sequence": 4294967293
      }
    ],
    "vout": [
      {
        "value": 0.0025,
        "n": 0,
        "scriptPubKey": {
          "asm": "OP_DUP OP_HASH160 7c58efc3e5a30cea3873e905e29ea4d9165d4d6d OP_EQUALVERIFY OP_CHECKSIG",
          "desc": "addr(mrrSdGDLiem8dTv97LSnZpcf9vz65sofqq)#6jj6p4an",
          "hex": "76a9147c58efc3e5a30cea3873e905e29ea4d9165d4d6d88ac",
          "address": "mrrSdGDLiem8dTv97LSnZpcf9vz65sofqq",
          "type": "pubkeyhash"
        }
      },
      {
        "value": 0.00149775,
        "n": 1,
        "scriptPubKey": {
          "asm": "OP_DUP OP_HASH160 8edb9697e77b2b7058d6b601c5d833e4011addfa OP_EQUALVERIFY OP_CHECKSIG",
          "desc": "addr(mtYKJZ3tz7DnnBzKzqu1A87nvbB9YxKSNv)#mn94ykmq",
          "hex": "76a9148edb9697e77b2b7058d6b601c5d833e4011addfa88ac",
          "address": "mtYKJZ3tz7DnnBzKzqu1A87nvbB9YxKSNv",
          "type": "pubkeyhash"
        }
      }
    ]
  },
  "tx_summary": {
    "txid": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
    "hash": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
    "version": 2,
    "size": 225,
    "vsize": 225,
    "weight": 900,
    "locktime": 0,
    "vin_count": 1,
    "vout_count": 2
  },
  "script_view": {
    "tx_summary": {
      "txid": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
      "hash": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
      "version": 2,
      "size": 225,
      "vsize": 225,
      "weight": 900,
      "locktime": 0,
      "vin_count": 1,
      "vout_count": 2
    },
    "inputs": [
      {
        "txid": "89a53f5b8b43846bf99f8869963c405751835a91734d064806471d800f378f05",
        "vout": 0,
        "sequence": 4294967293,
        "scriptsig_asm": "304402200d3cdf2926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a23[ALL] 027a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65",
        "scriptsig_hex": "47304402200d3cdf2926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a230121027a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65",
        "txinwitness": []
      }
    ],
    "outputs": [
      {
        "value": 0.0025,
        "n": 0,
        "scriptpubkey_type": "pubkeyhash",
        "scriptpubkey_asm": "OP_DUP OP_HASH160 7c58efc3e5a30cea3873e905e29ea4d9165d4d6d OP_EQUALVERIFY OP_CHECKSIG",
        "scriptpubkey_hex": "76a9147c58efc3e5a30cea3873e905e29ea4d9165d4d6d88ac",
        "address": "mrrSdGDLiem8dTv97LSnZpcf9vz65sofqq",
        "addresses": []
      },
```

SCRIPT VALIDATION EXPLANATION:

In a Legacy P2PKH transaction, the unlocking script (scriptSig / response) provides the user's cryptographic <Signature> and <Public Key>.

The locking script (scriptPubKey / challenge) validates this by running OP_DUP to duplicate the public key, and OP_HASH160 to hash it. It then pushes the destination address hash onto the stack and uses OP_EQUALVERIFY to ensure the hashes match. Finally, OP_CHECKSIG consumes the signature and public key to verify the mathematical proof of ownership.

DECODED TRANSACTION:

Scriptsig(unlocking script):


```
        "scriptSig": {
          "asm": "304402206c93c7c6a52d896664d7f21430729c34666886ddebffd74beb1ccc618e5cd9ec02205e93d0e814b067ce9f1d20e6ebe3378405e185f6bf58cb9517621add3a94073a[A
LL] 03c423f233f957f9144dc4303f0924fce9c56e409dc02185b92ebfdab2e15cfc7a",
          "hex": "47304402206c93c7c6a52d896664d7f21430729c34666886ddebffd74beb1ccc618e5cd9ec02205e93d0e814b067ce9f1d20e6ebe3378405e185f6bf58cb9517621add3a94073a
012103c423f233f957f9144dc4303f0924fce9c56e409dc02185b92ebfdab2e15cfc7a"
        },
        "sequence": 4294967293
```


Scriptpubkey(locking script):


```
    "scriptPubKey": {
      "asm": "OP_DUP OP_HASH160 dc5ae01a049516a1c1c72d913d890243869a8c1f OP_EQUALVERIFY OP_CHECKSIG",
      "desc": "addr(n1c5pPaBdJ7Njou41bQzUhLWtHDNvHYhDb)#ym4s6r3u",
      "hex": "76a914dc5ae01a049516a1c1c72d913d890243869a8c1f88ac",
      "address": "n1c5pPaBdJ7Njou41bQzUhLWtHDNvHYhDb",
      "type": "pubkeyhash"
    }
```


CHALLENGE-RESPONSE ANALYSIS:


```
==================== Legacy Challenge / Response Analysis ====================
{
  "mode": "legacy",
  "previous_txid": "89a53f5b8b43846bf99f0869963c405751835a91734d064806471d800f378f05",
  "spending_txid": "ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674",
  "challenge_script": {
    "source": "previous transaction output scriptPubKey",
    "asm": "OP_DUP OP_HASH160 dc5ae01a049516a1c1c72d913d890243869a8c1f OP_EQUALVERIFY OP_CHECKSIG",
    "hex": "76a914dc5ae01a049516a1c1c72d913d890243869a8c1f88ac",
    "type": "pubkeyhash",
    "address": "n1c5pPaBdJ7Njou41bQzUhLWtHDNvHYhDb"
  },
  "response_script": {
    "source": "spending transaction input scriptSig",
    "asm": "3044022008d3cd72926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a23[ALL] 02
7a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65",
    "hex": "473044022008d3cd72926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a23012102
7a487b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65"
  },
  "witness": [],
  "explanation": "For Legacy P2PKH, the challenge is the previous output's scriptPubKey (typically OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG). T
he response is the spending input's scriptSig, which typically pushes the signature and public key. Bitcoin validates the spend by executing scriptSig followed
by scriptPubKey; the pubKey hash must match and the signature must verify."
}
==============================================================================


=== btcdeb command for Legacy validation ===
btcdeb '3044022008d3cd72926bb7933437c0062f27be3f8b062d424fbc69d0760cdb77fa4c84f902205830451da401dcab83f154bc280530baf78766cac1e8900adfb461d8915c8a23[ALL] 027a48
7b807edb8f5ad6ceffa34644e5da8f493fc4c4dca4459143a4e999102e65 OP_DUP OP_HASH160 dc5ae01a049516a1c1c72d913d890243869a8c1f OP_EQUALVERIFY OP_CHECKSIG'
```

This block explains :

Challenge_Script→scriptPubkey

Response_script→scriptSig

FINAL WORKFLOW SCREENSHOT:

```
=== Legacy run completed successfully ===
A -> B txid: 89a53f5b8b43846bf99f0869963c405751835a91734d064806471d800f378f05
B -> C txid: ebe3be3abc08e8e153ad87d4a0e493e93e785bb29181703c3685147b2ec9a674
Artifacts saved under outputs/legacy/
```

PART-2: SEGWIT TRANSACTION

STEP-1:

Checking the status of the connection.

```
C:\Users\krant\OneDrive\Documents\Desktop\blockchain2\CS216-MerkleMinds-Bitcoin-Transaction>bitcoin-cli getblockchaininfo
{
  "chain": "regtest",
  "blocks": 108,
  "headers": 108,
  "bestblockhash": "1abf5dd271dd995ef2a3b9493eb232f65ee94cac677c8203cade03c30e095c20",
  "bits": "207fffff",
  "target": "7fffff0000000000000000000000000000000000000000000000000000000000",
  "difficulty": 4.656542373906925e-10,
  "time": 1773160381,
  "mediantime": 1773159852,
  "verificationprogress": 1,
  "initialblockdownload": false,
  "chainwork": "00000000000000000000000000000000000000000000000000000000000000da",
  "size_on_disk": 34614,
  "pruned": false,
  "warnings": [
  ]
}
```

STEP-2:Creating addresses and funding and creating transactions from A to B and B to C

```
"part": "Part 2 - P2SH-P2WPKH",
"wallet_name": "cs216wallet",
"addresses": {
  "A_prime": "2NBHcKHTSEZ9Woyf6stKxeisEk1PGEznn85",
  "B_prime": "2N6p11RVQs9RXxvNgeDKJ1zHQD6cHH6igvb",
  "C_prime": "2Mv5EQqdV7bU2pJt5c6EUMQayy1B4khYyHs",
  "address_type": "p2sh-segwit"
},
"funding": {
  "funding_txid": "4bd551a4459b7a1b0a2a15f94afd192dff5ec5d544563f6aa90ecdcba25456fb",
  "funded_address": "2NBHcKHTSEZ9Woyf6stKxeisEk1PGEznn85",
  "fund_amount": 0.01,
  "confirmation_blocks_mined": 1,
  "mining_address": "bcrt1qs2rkh7vct5xr7et745kr6yt8dzxlv8j3w6sdtr"
},
"tx_chain": {
  "A_prime_to_B_prime": {
    "txid": "cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba",
    "recipient_output_index_for_B_prime": 0,
    "saved_paths": {
      "full_artifact": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\A_prime_to_B_prime_cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba_full.json",
      "decoded_tx": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\A_prime_to_B_prime_cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba_decoded.json",
      "scripts": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\A_prime_to_B_prime_cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba_scripts.json",
      "summary": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\A_prime_to_B_prime_cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba_summary.json"
    }
  },
  "B_prime_to_C_prime": {
    "txid": "01f1732a1057346fd228c49ce8c453a2f400f5408372fb9a6ffc08babaea7960",
    "saved_paths": {
      "full_artifact": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\B_prime_to_C_prime_01f1732a1057346fd228c49ce8c453a2f400f5408372fb9a6ffc08babaea7960_full.json",
      "decoded_tx": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\B_prime_to_C_prime_01f1732a1057346fd228c49ce8c453a2f400f5408372fb9a6ffc08babaea7960_decoded.json",
      "scripts": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\B_prime_to_C_prime_01f1732a1057346fd228c49ce8c453a2f400f5408372fb9a6ffc08babaea7960_scripts.json",
      "summary": "C:\\Users\\krant\\OneDrive\\Documents\\Desktop\\blockchain2\\CS216-MerkleMinds-Bitcoin-Transaction\\outputs\\segwit\\B_prime_to_C_prime_01f1732a1057346fd228c49ce8c453a2f400f5408372fb9a6ffc08babaea7960_summary.json"
    }
  }
```


The generated addresses are P2SH-SegWit (P2SH-P2WPKH) addresses.These addresses wrap a SegWit witness program inside a P2SH script.(Sample saved transaction is shown below).


```
  "signed_tx_hex": "02000000000101fb5654a2cbcd0ea96a3f5644d5c55eff2d19fd4af9152a0a1b7a9b45a451d54b010000001716001 4dac397ba64308010ba4fd0d9016836cabaac84e
2fdffffff02801a06000000000017a91494cc86c6ad0d8f7b9ea2959e5626c86563819e6e871a2709000000000017a914284f801156a726a87aa0a4fe088392a5f3e8337c8702473044022037
ebf50df4ca690c0658424a0d4545d339e70ceefd0eaf9a83870bbadb0a3b5b02203ecc6a1823a3bfd77c5339b07b25e89103ad108c8b2dd93d8a43a87e6fde27bb012102d50b79ab6ce972398
3b60fffd3a3792075d81a8f71ce1c3d7c8a107e57dbdb9400000000",
  "txid": "cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba",
  "decoded_tx": {
    "txid": "cefdb68bb76479f30a2bb1c161007f0793b64c817ea50fec0d1273ada906d8ba",
    "hash": "6c2bf79edc0b900a8ce9e63c1461c0753553467df7c204a6909e12abb8da6181",
    "version": 2,
    "size": 247,
    "vsize": 166,
    "weight": 661,
    "locktime": 0,
    "vin": [
      {
        "txid": "4bd551a4459b7a1b0a2a15f94afd192dff5ec5d544563f6aa90ecdcba25456fb",
        "vout": 1,
        "scriptSig": {
          "asm": "0014dac397ba64308010ba4fd0d9016836cabaac84e2",
          "hex": "160014dac397ba64308010ba4fd0d9016836cabaac84e2"
        },
        "txinwitness": [
          "3044022037ebf50df4ca690c0658424a0d4545d339e70ceefd0eaf9a83870bbadb0a3b5b02203ecc6a1823a3bfd77c5339b07b25e89103ad108c8b2dd93d8a43a87e6fde27bb01
",
          "02d50b79ab6ce9723983b60fffd3a3792075d81a8f71ce1c3d7c8a107e57dbdb94"
        ],
        "sequence": 4294967293
      }
    ],
    "vout": [
      {
        "value": 0.004,
        "n": 0,
        "scriptPubKey": {
          "asm": "OP_HASH160 94cc86c6ad0d8f7b9ea2959e5626c86563819e6e OP_EQUAL",
          "desc": "addr(2N6p11RVQs9RXxvNgeDKJ1zHQD6cHH6igvb)#da5e45et",
          "hex": "a91494cc86c6ad0d8f7b9ea2959e5626c86563819e6e87",
          "address": "2N6p11RVQs9RXxvNgeDKJ1zHQD6cHH6igvb",
          "type": "scripthash"
        }
      },
      {
        "value": 0.00599834,
        "n": 1,
        "scriptPubKey": {
          "asm": "OP_HASH160 284f801156a726a87aa0a4fe088392a5f3e8337c OP_EQUAL",
          "desc": "addr(2MvvNFjS1bsesAUXaLftiYAZQvc7LcfsMEG)#mgy07h4r",
          "hex": "a914284f801156a726a87aa0a4fe088392a5f3e8337c87",
          "address": "2MvvNFjS1bsesAUXaLftiYAZQvc7LcfsMEG",
          "type": "scripthash"
        }
```


The utxo from A'->B' is used as input for B'->C'

STEP-3: Script Validation Explanation:

In a **SegWit P2SH-P2WPKH transaction**, the unlocking data is moved to the **witness field**, which contains the user's cryptographic <Signature> and <Public Key>. The scriptSig only

holds the **SegWit redeem script**, which references the witness program.

The locking script (scriptPubKey / challenge) checks the ownership of the funds by hashing the provided public key using **OP_HASH160** and comparing it with the stored public key hash. If the hashes match, the script proceeds to **OP_CHECKSIG**, which verifies that the signature corresponds to the provided public key.

If the signature is valid and the hashes match, the script evaluation succeeds, proving that the spender has the correct private key and therefore the authority to spend the transaction output.

STEP-4 BTCDEB COMMAND:

```
Run this command for btcdeb debugging:

btcdeb '[304402200183d0735b3a9d25212277c37e9cf3abed302b6fd28b57e157689456e254477602200e3814487d2e8cdce622275fe76b36c90a23b4c8f6b77ff3679b38b03def
cbb801 029d3c3e1515ba0c78d7828d70826e8d4dcb47f6bc5586c35223a6a11a4b2b7473]' 'OP_DUP OP_HASH160 444a86b3439c8b23c9b53a22fd19345de2dc53ae OP_EQUALV
ERIFY OP_CHECKSIG'
```

btcdeb '[signature pubkey]'

'OP_DUP OP_HASH160 ... OP_EQUALVERIFY OP_CHECKSIG'

The above command shows a structure of the btcdeb command generated by the code to test the given transaction.

The btcdeb command is used to debug and validate the execution of

Bitcoin scripts. It allows step-by-step verification of the scriptSig,

scriptPubKey and witness stack during transaction validation.

STEP-5: FINAL SUMMARY OF ENTIRE CODE:

```
=================== SegWit Part 3 Metrics ===================
{
  "comparison_rows": [
    {
      "label": "A_prime_to_B_prime",
      "tx_type": "p2sh-p2wpkh",
      "size": 247,
      "vsize": 166,
      "weight": 661,
      "fee_savings_percent": 0.0
    },
    {
      "label": "B_prime_to_C_prime",
      "tx_type": "p2sh-p2wpkh",
      "size": 247,
      "vsize": 166,
      "weight": 661,
      "fee_savings_percent": 0.0
    }
  ],
  "average_metrics": {
    "avg_size": 247.0,
    "avg_vsize": 166.0,
    "avg_weight": 661.0
  },
  "witness_discount_examples": {
    "A_prime_to_B_prime": {
      "size": 247,
      "vsize": 166,
      "weight": 661,
      "formula": {
        "weight": "(non-witness bytes \u00d7 4) + witness bytes",
        "vsize": "ceil(weight / 4)"
      },
      "explanation": "SegWit moves signatures to txinwitness. Witness bytes are discounted in weight calculation, so vsize is often significantly smaller
than raw size. Because fees are paid using vsize, SegWit transactions are cheaper."
    },
    "B_prime_to_C_prime": {
      "size": 247,
      "vsize": 166,
      "weight": 661,
      "formula": {
        "weight": "(non-witness bytes \u00d7 4) + witness bytes",
        "vsize": "ceil(weight / 4)"
      },
      "explanation": "SegWit moves signatures to txinwitness. Witness bytes are discounted in weight calculation, so vsize is often significantly smaller
than raw size. Because fees are paid using vsize, SegWit transactions are cheaper."
    }
  }
}
============================================================
```

SUMMARY:

SegWit reduces transaction size because the witness data is discounted when computing the virtual size (vsize), which is calculated as weight / 4.

We have successfully implemented the requirements and also saved the transactions as json files into the code base which represent the transactions being done in code. We have also saved json files for the btcdeb (bitcoin debugger) commands into the codebase as well. The requirements showing funding, transactions from A to B and then B to C and script verification and btcdeb are all done.

PART-3:

Analysis and Comparison of Legacy (P2PKH) and SegWit (P2SH-P2WPKH) Transactions

(P2SH-P2WPKH) Transactions

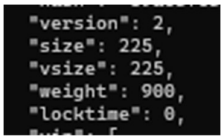
Transaction Size Comparison

The transactions we have used are at some stage of development and we have made a few changes and have rerun the code which may cause the data in the report here not similar to the transaction data in the repository but the code structure of differences and comparisons remain the same.

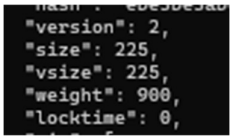
The transactions generated in **Part-1 (Legacy P2PKH)** and **Part-2 (SegWit P2SH-P2WPKH)** were analyzed using decoderawtransaction to extract their size related metrics.

Part 1 – Legacy P2PKH Transaction Sizes

Transaction A->B:



TRANSACTION B->C:



Part 2 – SegWit P2SH-P2WPKH Transaction Sizes

TRANSACTION A'->B':


```
"hash": "6c2bf79edc0b900a8ce9e63c1461c0753553467df7c204a6909e12abb8da6181",
"version": 2,
"size": 247,
"vsize": 166,
"weight": 661,
"locktime": 0,
```


TRANSACTION B'->C':


```
"version": 2,
"size": 247,
"vsize": 166,
"weight": 661,
"locktime": 0,
```


Although the total byte size of the SegWit transaction appears slightly larger due to additional structural fields (marker, flag, witness stack), the **virtual size (vsize)** and **weight units** are significantly lower. This occurs because the **witness data is discounted when calculating transaction weight** — witness bytes count as 1 weight unit each, whereas non-witness bytes count as 4 weight units each.

Script Structure Comparision:

Legacy P2PKH Transaction Scripts

In Legacy transactions, both the unlocking and locking scripts are executed together in the same validation context. The scripts are concatenated and run sequentially on the stack.

Scriptsig(unlocking script):


```
"scriptSig": {
  "asm": "304402206c93c7c6a52d896664d7f21430729c34666886ddebffd74beb1ccc618e5cd9ec02285e93d8e814b067ce9f1d20e6ebe3378405e185f6bf58cb9517621add3a90073a[ALL] 03c423f233f957f9144dc4383f0924fce9c56e489dc02185b92ebfdab2e15cfc7a",
  "hex": "47304402206c93c7c6a52d896664d7f21430729c34666886ddebffd74beb1ccc618e5cd9ec02285e93d8e814b067ce9f1d20e6ebe3378405e185f6bf58cb9517621add3a90073a012103c423f233f957f9144dc4383f0924fce9c56e489dc02185b92ebfdab2e15cfc7a"
},
"sequence": 4294967293
```

Scriptpubkey(locking script):

```
"scriptPubKey": {
  "asm": "OP_DUP OP_HASH160 dc5ae01a049516a1c1c72d913d890243869a8c1f OP_EQUALVERIFY OP_CHECKSIG",
  "desc": "addr(n1c5pPaBdJ7Njou41bQzUhLWtHDNvHYhDb)#ym4s6r3u",
  "hex": "76a914dc5ae01a049516a1c1c72d913d890243869a8c1f88ac",
  "address": "n1c5pPaBdJ7Njou41bQzUhLWtHDNvHYhDb",
  "type": "pubkeyhash"
}
```


The **challenge script** uses OP_DUP OP_HASH160 to duplicate the provided public key and hash it with HASH160. OP_EQUALVERIFY checks that the computed hash matches the stored public key hash. Finally OP_CHECKSIG verifies the cryptographic signature, proving ownership of the funds

SegWit P2SH-P2WPKH Transaction Scripts

In SegWit transactions, the signature and public key are separated from the main transaction data and stored in the **witness field**. The scriptSig only holds the SegWit redeem script reference, keeping the main transaction body compact

Locking Script:


```
{
  "value": 0.00149834,
  "n": 1,
  "scriptPubKey": {
    "asm": "OP_HASH160 c6a219953e0bdeafcac8fba90d90ddf219df38d7 OP_EQUAL",
    "desc": "addr(2NBMVyAejqTzbsrcFxqpDT1BVMTqNXSf8F7)#fqucg02n",
    "hex": "a914c6a219953e0bdeafcac8fba90d90ddf219df38d787",
    "address": "2NBMVyAejqTzbsrcFxqpDT1BVMTqNXSf8F7",
    "type": "scripthash"
  }
}
]
```


Scriptsig:

```
  "txid": "4bd551a4459b7a1b0a2a15f94afd192dff5ec5d544563f6aa90ecdcba25456fb",
  "vout": 1,
  "scriptSig": {
    "asm": "0014dac397ba64308010ba4fd0d9016836cabaac84e2",
    "hex": "160014dac397ba64308010ba4fd0d9016836cabaac84e2"
  },
  "txinwitness": [
    "3044022037ebf50df4ca690c0658424a0d4545d339e70ceefd0eaf9a83870bbadb0a3b5b02203ecc6a1823a3bfd77c5339b07b25e89103ad108c8b2dd93d8a43a87e6fde27bb01

    "02d50b79ab6ce9723983b60fffd3a3792075d81a8f71ce1c3d7c8a107e57dbdb94"
  ],
  "sequence": 4294967293
}
```


The redeem script in the scriptSig references the SegWit witness program, and the actual unlocking data is provided through the witness stack. During validation, the public key is hashed and compared with the stored hash, and the signature is verified using OP_CHECKSIG.

Why SegWit Transactions are More Efficient

Segregated Witness improves transaction efficiency by separating signature data from the main transaction structure. The weight formula for a SegWit transaction is:

```
Weight  =  (non-witness bytes  x  4)  +  (witness bytes  x  1)

Legacy  P2PKH:        225 bytes x 4  =  900  weight units
                      vSize = ceil(900 / 4) = 225 vbytes

SegWit  P2SH-P2WPKH:
  Non-witness part: ~110 bytes x 4  =  440
  Witness part:     ~107 bytes x 1  =  107
  Remaining fields: ~28  bytes x 4  =  114
  Total weight                      =  661  weight units
  vSize = ceil(661 / 4)             =  166  vbytes   ← 26% smaller
```

ADVANTAGES OF SEGWIT:

Reduced virtual transaction size (vsize) — 166 vs 225 vbytes (~26% reduction), allowing more transactions to fit within a block.

Lower effective transaction fees — fees are calculated based on vbytes rather than raw bytes, so SegWit users pay proportionally less.

Elimination of transaction malleability — signature data is moved outside the transaction ID calculation, improving reliability for higher-layer protocols such as the Lightning Network. Furthermore, the witness program includes a version byte, paving the way for easier future network upgrades (like Taproot).

Improved block capacity — the effective block capacity increases from 1 MB to up to 4 MB in weight units (~1.7–2 MB in practice), enabling the network to process more transactions per block.

Backward-compatible deployment — P2SH-P2WPKH wraps the SegWit witness program inside a familiar P2SH envelope, ensuring compatibility with older wallets that do not natively support SegWit.

CONCLUSION:

In this project we explored the structure and execution of Bitcoin transactions using both **Legacy (P2PKH)** and **SegWit (P2SH-P2WPKH)** address formats. By interacting with a Bitcoin node in Regtest mode and creating transactions programmatically through Python, we were able to observe how the **UTXO model** functions and how transactions are validated using Bitcoin's scripting system.

In Part 1, we implemented a Legacy transaction chain where funds were transferred from address **A to B and then from B to C**. Through this process, we observed how a transaction output becomes a **UTXO (Unspent Transaction Output)** that can later be spent in subsequent transactions. The analysis of the **scriptPubKey (locking script)** and **scriptSig (unlocking script)** demonstrated how the Bitcoin network verifies ownership through digital signatures and public key hashes.

In Part 2, we repeated the same transaction workflow using **SegWit addresses**. The SegWit implementation introduced the concept of separating signature data into the **witness structure**, which moves signature and public key information outside the main transaction body. This separation reduces the effective transaction weight and mitigates issues such as transaction malleability.

Finally, the comparison between Legacy and SegWit transactions highlighted the improvements introduced by SegWit. Although both transaction types follow the same fundamental UTXO model, SegWit transactions are more efficient due to reduced **virtual size (vsize)** and improved fee efficiency. By moving signature data to the witness field, SegWit transactions lower the effective block weight and increase overall network scalability.

Overall, this project provided practical insight into how Bitcoin transactions are created, validated, and propagated across the network. It demonstrated the importance of scripting mechanisms in securing transactions and illustrated how protocol improvements such as SegWit enhance performance and efficiency within the Bitcoin ecosystem.